Multi-Robot Coordinated Motion Planner Ensuring Line-of-Sight Connectivity

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***Abstract*— The use of multi-robot teams in communication-denied environments poses a challenge to coordinated operations due to signal degradation and the lack of a direct line-of-sight (LOS) caused by obstacles. This paper introduces a centralized, coordinated motion planner designed to establish and maintain a communication chain by ensuring uninterrupted LOS connectivity. Our methodology leverages a visibility graph and a custom path-finding algorithm that optimizes for both path length and the number of required relay agents. The system then generates a deterministic initial order of robots movement through the path and incorporates a reactive re-planning mechanism to handle environmental interruptions, such as signal blockages or dynamic obstacles, reorganizing the robots to a new calculated path. The use of these two order of movements algorithms provides a flexible framework that allows a choice between a computationally intensive adaptive repair for a direct solution, or a simpler, faster full redeployment for more complex scenarios. The approach was validated in a simulation environment using ROS 2, Gazebo, and Open-RMF. Results demonstrate the fleet's ability to execute a nominal deployment, detect and react to blockages by reconfiguring its formation, and circumnavigate obstacles while preserving LOS connectivity at every step, confirming the viability of the strategy for multi-robots communication networks. The comparison of the adaptive re-planing method against the deterministic one presented a gain of 12.21% of the total distance traveled by all the robots and 22.73% in the number of movement to reach the goal.**

I. INTRODUCTION

Despite rapid advances in autonomous navigation and task planning, the deployment of service mobile robots remains constrained by the availability and reliability of communication infrastructure. In many real-world settings, from disaster-response sites and remote agricultural fields, to industrial warehouses, robots must operate in areas that do not have proper connection, causing problems for tasks such as coordinated multi-agent behaviours, and even basic supervisory control [1]. This communication gap not only degrades mission performance but also raises safety and accountability concerns. Consequently, there is a need for resilient on-board communication strategies and decision-making architectures that allow mobile robots to self-organize.

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One mitigation strategy is to leverage the robots themselves as mobile relay nodes, forming an ad-hoc mesh network in which each platform shares its radio link with nearby peers to propagate data toward a base station [2]. Such topologies can dramatically extend operational range without costly fixed infrastructure. Moreover, the mesh network can reconfigure as robots move, preserving connectivity in dynamic or partially obstructed environments and facilitating cooperative tasks like distributed mapping and inspections.

However, these benefits come at the price of added protocol complexity, multihop latency, and higher energy draw, since every relay must allocate computational and power resources to packet forwarding. The network's throughput is further constrained by shared medium contention and cumulative interference, while route maintenance overhead grows with fleet size and mobility. Finally, security and quality-of-service guarantees become harder to enforce when each robot doubles as both application agent and network router, necessitating robust encryption, trust management, and real-time traffic prioritization mechanisms.

Beyond protocol design, another important challenge for mesh fleets is the spatial distribution of individual robots: maintaining sufficient robot proximity and ideally line-of-sight (LOS) so that signal strength and link quality remain above a minimal threshold. Poor placement can create sparse regions where hop counts surge, latency accumulates, and packets are lost, potentially leading to the complete loss of communication. Strategically arranging robots to preserve LOS whenever feasible, mitigates this problem, as direct visual paths decrease multipath fading and obstruction losses [3], [4]. However, achieving and sustaining such geometries in cluttered, dynamic environments demands continuous topology monitoring and motion planning [5], [6], [7].

Orchestrating robot positions to maintain network geometries is a challenge addressable by using a fleet management framework, such as the open-source Open-RMF. Such a system provides the essential tools for continuous topology monitoring and centralized motion planning, scheduling collision-free paths for multiple agents to sustain optimal formations and preserve LOS links. Furthermore, while the framework allows the robot to execute repositioning commands, it does not inherently contain the high-level, connectivity-aware logic to decide which robot should move where. We still need to develop that strategic layer to command the fleet manager.

To address the robot distribution problem, our paper introduces a coordinated multi-robot motion planner that explicitly couples task objectives with connectivity constraints.

Formulated as an optimization problem over a dynamically updated visibility graph, the planner prioritizes relay repositioning ahead of task-level actions whenever link margins degrade or a dynamic obstacle is found, thereby preventing topology fragmentation before it occurs. By embedding these communication guarantees and adaptability to dynamic obstacles directly into the motion planning, the proposed approach enables the fleet to adaptively re-lace its LOS chain in cluttered, non-stationary environments while still advancing toward spatial goals.

Then, we compute an ordered sequence of movements for each platform, given by two possible approaches: a deterministic deployment strategy and a reactive re-planning one, such that (i) every robot designated as a relay maintains uninterrupted LOS with at least one other member of the fleet, (ii) the aggregate topology always forms a connected backbone to the command station, with the the communication-aware motion method implemented in a fleet management framework.

We also consider a strategy of functional heterogeneity to balance mission capability with economic and operational constraints. Thus, a single, high-capability robot is designated as the primary or lead agent, uniquely outfitted with the specialized sensors or actuators required for a specific task, such as a high-resolution camera for inspection or a manipulator arm for interaction. The remainder of the fleet consists of simpler, lower-cost platforms whose primary function is to act as relay nodes or support units. These support robots form a distributed infrastructure, extending the communication range of the primary agent, providing logistical support, or creating a sensor network over a wide area. This division allows the team to achieve complex objectives without the prohibitive cost of deploying a homogeneous fleet of fully-equipped, and therefore expensive, robots.

The rest of this paper is organized in sections as follows. Section II show the use of communication-aware fleet management systems. Then, it follows through section III and IV, where the methodology for our approach is presented, including the planner and the order of movement methods and its implementations respectively. Section V show the results achieved and the comparison of our two deployment order methods in an simulated environment. Finally, Section VI present the concluding remarks and future works.

II. Related works

The deployment of mobile robots in challenging environments, such as subterranean scenarios, is often constrained by communication infrastructure due to limited line-of-sight and complex signal propagation [8], [4]. While radio signal quality has been primarily used for robot localization [9], [10], it also serves to take navigation decisions based on proximity to access points [11] and for antenna placement optimization.

Several works have explored using communication signal strength for multi-robot control strategies, enabling robots to reorganize for improved connectivity [5]. Path planning algorithms have also incorporated communication quality as a key metric, aiming to find paths that prioritize better signal strength [12], [6]. Addressing unknown environments, signal prediction strategies, such as PropEM-L [13], have been developed to estimate signal intensity adaptively.

A significant body of work addressing constrained communication in complex environments stems from the DARPA Subterranean Challenge [14]. Teams in this competition extensively utilized mesh networks formed by heterogeneous autonomous robot systems to optimize communication with external human operators. For instance, the MARBLE team focused on transport prioritization and a custom UDP-based mesh solution for high autonomy systems [2]. In contrast, teams like NeBula [15] and CSIRO [16] proposed solutions less dependent on autonomy, employing deployable radios to form a communication backbone. Team CERBERUS [17] further developed a heterogeneous mesh combining both deployable radios and mobile robots to establish Wi-Fi connectivity.

Regarding the line-of-sight problem in multi-robot systems, existing researches employ different methods. A centralized approach is to use Mixed-Integer Programming to compute feasible paths that meet safety constraints, including obstacle avoidance, connectivity, and LOS [18]. Additional research leverages a central server to aggregate and disseminate obstacle data among robots, dynamically replanning trajectories to minimize collision risks [19]. In parallel, decentralized controllers can maintain LoS by minimally filtering nominal motions with probabilistic control-barrier functions to guarantee connectivity under pose uncertainty [20]. Alternatively, Perception-driven distributed methods can be used to infer LoS from live point clouds in unknown environments and coordinate robots to keep the communication graph connected by sustaining a positive Fiedler eigenvalue [21].

Building upon these multi-robot coordination strategies, high-level fleet management systems such as the open-source Robotics Middleware Framework (Open-RMF) have been used for task automation with heterogeneous mobile robot fleets. Their effectiveness has been shown in crowded hospital environments, coordinating task distribution and transporting time-critical samples [22]. Open-RMF has also been integrated with methodologies like Control Barrier Functions (CBFs) to enhance safety for human-robot navigation in warehouse scenarios [23]. However, there is still no method for connection-aware planning available.

In this sense, different from the works incorporating path planning methods into robotic fleet management systems, which generally addresses high-level task coordination and obstacle-based path planning, this work concentrates on the generation of a path planning strategy, involving the underlying motion procedures required to create efficient paths that maintain wireless network connectivity within a fleet of robots.

III. Methodology

The primary objective of this paper is to implement a multi-robot coordination system capable of forming and

maintaining a connected communication chain from a base station to a target. The core challenge addressed is ensuring uninterrupted end-to-end network connectivity throughout all operational phases, including initial deployment and adaptive reconfiguration in response to environmental or communication obstacles. The system guarantees that at no point during an agent's transit does it, or any other agent, become disconnected from the primary communication backbone originating at the base. This connectivity maintenance is the fundamental constraint governing all planning and execution logic.

The implemented algorithm employs a centralized, iterative approach to manage the multi-agent team. To prevent suboptimal or chaotic formations, a heuristic control policy termed Ordered Progression is enforced. This policy ensures that the robotic chain is constructed in a structured, sequential manner from the base towards the target. The Ordered Progression rule prevents agents from occupying nodes significantly ahead of the contiguous formation front. This method guarantees that forward positions are reserved for higher-priority agents, specifically the lead agent tasked with reaching the final goal, thus maintaining a clear and logical path for its advance.

To validate the safety of each potential move, this work utilizes a trajectory sampling method. Despise collision-free path guaranteed by a visibility graph is sufficient to ensure continuous communication among nodes, the route between nodes also has to be validated. Thus, this method discretizes any proposed straight-line trajectory into an intermediate sample point. For an agent to be cleared for transit, it must be demonstrated that from this sample point, a direct line-of-sight to at least one static agent in the existing chain can be established. A trajectory is only approved if this connectivity check passes for all sample points, effectively mitigating the risk of transient network disconnections in areas of sparse visibility.

A. Visibility Graph

Our methodology starts with the generation of a base visibility graph, $G = (V, E)$, from a 2D occupancy grid map, M. The map is processed to identify features for graph vertices. Obstacle boundaries are extracted using a Canny edge detector, followed by morphological operations to create a configuration space, $\mathcal{C}_{free}$. Within this space, vertices are identified using a corner detection algorithm. To mitigate redundancy from feature points, the detected corners are clustered using a Density-based clustering algorithm with a distance threshold ϵ. The centroid of each cluster is designated as a vertex, forming the vertex set V. The edge set E is constructed by connecting any two vertices, $(v_i, v_j) \in V \times V$, if the line segment $\overline{v_i v_j}$ lies within $\mathcal{C}_{free}$. This visibility check is implemented by discretizing the segment and verifying that the number of points intersecting the obstacle space, $\mathcal{C}_{obs}$, does not exceed a tolerance threshold, ensuring a line of sight between the vertices.

To account for environments, the system incorporates a mechanism to update the graph in response to obstacles, physical or communication ones, like the sudden drop of the signal strength due to unpredictable interferences. Upon detecting an obstacle at a point $p_{obs} \in \mathcal{C}_{free}$, the algorithm identifies the corridor being blocked. This is achieved by leveraging a Euclidean Distance Transform of the map, $D(p)$, and its gradient field, $\nabla D(p)$. A point on an obstacle boundary, p_{c1}, is found by stepping from p_{obs} in the direction of $-\nabla D(p)$. A boundary point of the corridor, p_{c2}, is located by traversing from p_{c1} along the normal vector, derived from $\nabla D(p_{obs})$. The line segment, $S_{block} = \overline{p_{c1} p_{c2}}$, represents the passage. The system then identifies a set of edges for removal, containing edges from the graph that intersect this segment:

$$E_{int} = \{e \in E \mid e \cap S_{block} \neq \emptyset\}. \tag{1}$$

A refinement is applied to prevent the robot from removing path segments behind it resulting in the possibility of not beeing able to return. This is achieved by filtering the edges E_{int} based on the robot's path, P_{path}. For each edge $e_{ij} \in E_{int}$, a direction vector $\vec{d}_{ij}$ is determined relative to the obstacle's position. This vector is compared against the robot's path direction vector, $\vec{d}_{path}$, which is tangent to the path segment near p_{obs}. An edge is marked for removal if it points in the robot's direction of travel, a condition satisfied when their dot product is positive, where τ is a positive threshold:

$$\vec{d}_{path} \cdot \vec{d}_{ij} > \tau. \tag{2}$$

This yields the set of edges to remove, $E_{rem} \subseteq E_{int}$. The graph is reconstructed as $G' = (V', E')$, where the vertex set is expanded to include the start, goal, and obstacle locations:

$$V' = V \cup \{p_{start}, p_{goal}, p_{obs}\}. \tag{3}$$

The edge set E' is formed by re-evaluating visibility for pairs in V' and excluding the edges in E_{rem}.

B. Path planning

Prior to the physical deployment of any robotic agent, the system first determines an optimal static path from the base station p_0 to the goal destination. This is achieved by performing a shortest path search on the pre-computed visibility graph balancing the number of nodes and distance. The fundamental constraint of the system is to form an unbroken communication chain. For a path $P = (p_0, p_1, \ldots, p_{k-1})$ consisting of k nodes to be viable, it requires a deployment of $k-1$ agents, $\{r_1, \ldots, r_{k-1}\}$, such that the final assigned position of agent r_i is the node p_i. In this distribution, the lead robot (r_1) is assigned as the one that have to reach the goal and the others act as relay nodes in the path.

To find a path that is efficient in both distance and the number of required agents, a custom cost function is minimized. The total cost $C(P)$ of any given path is the sum of its geometric edge lengths and a penalty term, λ, for each node traversed. For a path P with $k-1$ segments, the cost is formally defined as:

$$C(P) = \sum_{i=1}^{k-1} w(p_{i-1}, p_i) + (k-1) \cdot \lambda. \quad (4)$$

Here, $w(p_{i-1}, p_i)$ is the Euclidean distance for the edge between nodes p_{i-1} and p_i. A Dijkstra-based shortest path algorithm is employed to find the path P^* that minimizes this cost function over the set of all possible paths $\mathcal{P}$. The resulting path P^* represents the ideal route for the robot chain, as the λ parameter biases the search towards paths that require the fewest nodes—and therefore the fewest robots to connect the base to the goal.

C. Deployment Strategy: Deterministic Order of Movement

Considering the plan of the initial deployment onto the optimal path generated in the previous stage, the primary goal of this function is to generate a deterministic, open-loop sequence of commands to position the $n-1$ agents. This establishes the initial communication backbone required for subsequent operations. The algorithm assumes all agents are initially co-located at the path's origin, p_0, and generates a complete, pre-calculated plan to achieve the final deployed configuration. The algorithm's structure is based on a fixed combinatorial pattern that dictates the movement of each agent. For the given path $P^* = \{p_0, p_1, \ldots, p_{n-1}\}$ with n nodes, the total number of individual agent movements, M, required to complete the deployment is determined by the $(n-1)$-th triangular number, given by the formula:

$$M = \frac{(n-1)n}{2} = \sum_{i=1}^{n-1} i. \quad (5)$$

The sequence of moves is generated by iterating M times, systematically updating the index of the moving agent and its target node. The algorithm selects agents to move according to its rigid, pre-defined sequence (e.g., r_1, r_2, r_1, r_3, r_2,...). From a computational standpoint, the deterministic deployment algorithm exhibits a time complexity of $O(n^2)$, as the number of generated movements is quadratically proportional to the number of nodes in the path.

D. Replaned order of movement

In response to a path obstruction, the planner executes a dynamic reconfiguration. A new optimal path, P_{new}, is computed on an updated graph G' using the same balanced cost function. The system then generates a new movement sequence, $S_{reconfig}$, through an iterative heuristic search, shown in Algorithm 1. At each step, a robot r_i and a target vertex v_{target} are selected based on a decision hierarchy. The primary heuristic is progression: a robot located on P_{new} attempts to advance to the subsequent vertex. If a robot is not on P_{new}, an acquisition heuristic is employed. This heuristic directs the robot to a neighboring vertex in G' that offers visibility to the most distant reachable vertex along P_{new}, maximizing the extension of the chain toward the goal. The node selected is the one with minimum distance to travel to P_{new} where first, we Identify all reachable neighbors and evaluate each one based on a cost function that (i) Maximizes progress towards P_{target}, (ii) Minimizes travel distance, and (iii) Preserves a line-of-sight to the existing robot chain.

Algorithm 1 Coordinated adaptive movement

Require: Robot Positions P, Target Path P_{target}
Ensure: A motion plan

```
while goal is not reached do
    for each robot r ∈ P do
        Generate a move p_cand for r towards P_target
        isSafe ← ValidateMove(r, p_cand, P)
        if isSafe then
            Execute move for r to p_cand
            Update P
        else
            Stay in current position
        end if
    end for
end while
```

Each proposed robot movement is subject to validation against two constraints, presented in Algorithm 2 to ensure formation integrity and prevent deadlocks. First, a connectivity constraint is verified by simulating the moving robot at the midpoint of its trajectory. An instantaneous visibility graph, G_{temp}, is constructed using the positions of all robots, and the move is authorized only if all robots remain connected to the base vertex within G_{temp}. To validate the topological integrity of the multi-robot formation, the system employs a connectivity check. A Breadth-First Search (BFS) algorithm originates from the base station node to identify the set of all reachable vertices within the graph. The formation's connectivity is confirmed if and only if the set of vertices currently occupied by robots is a subset of the set of vertices reachable from the base.

Algorithm 2 Validate Move

Require: A robot r, its $p_{current}$ and $p_{candidate}$, all robot positions P, Target Path P_{target}
Ensure: A boolean decision

```
if the move allows r to go over unoccupied nodes that should be filled first then
    return false                    ▷ Enforce sequential formation
end if
if the move blocks r_1 immediate advance then
    return false                    ▷ Prioritize the lead robot
end if
p_mid ← The midpoint of p_current to p_candidate
P_sim ← A simulated map state, with robot r at p_mid
G_temp ← A temporary visibility graph built from P_sim
is_connected ← Check if G_temp connects to the Base
if not is_connected then
    return false                    ▷ Move rejected
end if
return true                         ▷ Candidate move passes both checks
```

Second, a deadlock avoidance policy is enforced to man-

age traffic flow along the path. A robot at vertex $P_{new}[j]$ is prevented from advancing to $P_{new}[j+1]$ if the set of vertices ahead, $V_{future} = \{P_{new}[j+1], \ldots, P_{new}[k]\}$, is saturated with other robots. This policy ensures a vacant vertex is available for expansion, with specific rules granting priority to a designated lead robot (R1) to guarantee forward progress of the entire formation.

Our heuristic algorithm guarantees polynomial-time performance, ensuring a computationally tractable method for generating a feasible robot movement sequence, prioritizing finding a working solution quickly, which is crucial for real-time applications, but it does not guarantee the solution is optimal. In contrast, a formal optimization approach would formulate the task as a Mixed-Integer Linear Program (MILP) [24] to find a probable best solution. However, solving a MILP is an NP-hard problem, meaning the time required to find the optimal solution can grow exponentially with the problem's complexity, making it potentially infeasible for large-scale or time-sensitive scenarios.

The computational cost for generating a dynamic reconfiguration plan using the new order method is substantially higher than the cost of generating an deterministic initial deployment plan. The initial order method follows a deterministic pattern with a low-degree polynomial cost that scales quadratically with the path length, making it computationally inexpensive. In contrast, the new order method executes a complex heuristic search, where each step requires computationally intensive validation checks for connectivity and deadlock avoidance, resulting in a high-degree polynomial cost that is sensitive to the number of robots and the graph's complexity. For large-scale problems involving many robots or a dense visibility graph, this high cost may be prohibitive. In such cases, a more tractable alternative strategy can be employed: first, generate the simple reverse sequence to retract all robots along their initial path to the base, and then execute the computationally cheap initial deployment algorithm along the newly computed clear path.

IV. Implementation

For the implementation, we utilized the Robot Operating System (ROS) as the central middleware for managing communication between the perception and planning nodes. This includes map and path topics. The visibility graph is initialized from the 2D occupancy grid and uses OpenCV for edge extraction (Canny) and corner detection. The resulting keypoints are clustered using DBSCAN, which merges dense groups into single vertices. This produces the base set of nodes and edges of the visibility graph.

After initialization, the graph is not rebuilt at every change. When a new occupancy grid with the same metadata (size, resolution, origin) arrives, we update the graph incrementally and compute a region of interest (ROI) from the cells that changed occupancy, re-evaluating only the edges that intersect this ROI. Unchanged edges are kept, and a full rebuild is performed only if the map metadata changes.

Rerouting is triggered by events that modify the valid connectivity of the current graph. There are two cases. (i) Map updates: Once ROI revalidation finishes, the updated graph is published, and the planner re-optimizes paths. (ii) Local interference/obstacle: Upon an obstacle event, we apply pruning without reconstructing the graph and using the methodology stated in Section III-A, removing the edges of the blocked corridor and leaving only those that allow the robot to retreat if needed.

For the planner, LoS connectivity is enforced at every incremental motion step. Before any agent advances, each candidate displacement is evaluated using the rules of Algorithm 2. Global reachability to the base is required for all agents. In parallel, static graph connectivity and occupancy/progression constraints are verified. These steps preclude deadlocks or formations that would compromise LoS. If a locally feasible avoidance maneuver would entail a transient violation of the connectivity requirement (for example, due to a dynamic obstacle), the action is rejected rather than tolerated. The system adapts by replanning: the visibility graph is reprocessed, a new global route is computed, and the execution schedule is regenerated under the same LoS and progression safeguards. Finally, if the number of required support nodes along the route exceeds the available agents, the planner declares the problem infeasible and halts.

Our planner is deliberately centralized. Its applicability follows from a design choice that couples motion feasibility with communication feasibility. At execution time, LoS connectivity is enforced at every incremental motion step. Candidate displacements are admitted only if all agents remain reachable from the base through a visibility chain over the current occupancy model. Consequently, the central node can reliably disseminate a global route and an ordered execution schedule as short, discrete commands (such as advance permissions).

Agents can uplink newly sensed changes to the same node without risking loss of contact. When local sensing reveals a new obstacle or blockage, the visibility graph is reprocessed on the updated map. The schedule is regenerated. If any move jeopardizes the LoS chain, and with it the communication link, the action is not performed. Instead, a replan is triggered. In this way, the system's characterization ensure communication requirements and avoids configurations that would force decentralized operation.

A. Multi-Robot Execution with Open-RMF

The multi-robot fleet execution is orchestrated by the Open-Robotics Middleware Framework (Open-RMF) on ROS 2 Jazzy, simulated in Gazebo Harmonic, using Scout Mini, from Agilex Robotics (Fig.1), and the map depicted in Fig. 2. The movement sequence for each robotis provided as input to the Open-RMF system via `bash` commands that invoke the `dispatch_go_to_place` task[1]. When an interference is flagged by the path planning system,corrective commands are sent to RMF. A `dispatch_go_to_place` command re-assigns the agent to its last successfully traversed node, while a concurrent `cancel_task` command interrupts the execution of the now-invalid movement.

[1]https://github.com/open-rmf/rmf_demos

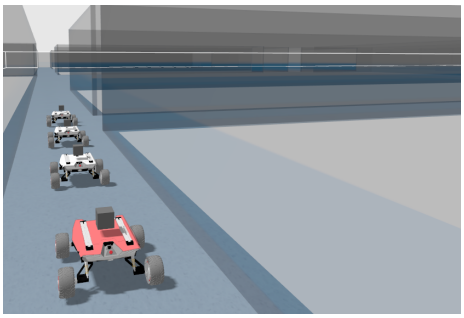


Fig. 1: Scout Mini fleet in Gazebo scene to simulate our deploy system. The leader robot is depicted in red.

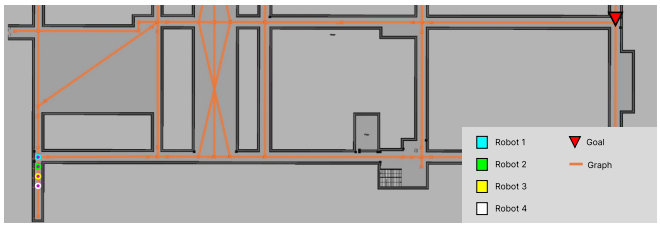


Fig. 2: Initial state of the simulation, showing the four robots (lower-left), the goal (top-right), and the RMF graph (orange).

V. Results

To test our approach, we used a occupancy grid map of the second floor of Engineering Building of the Federal University of Minas Gerais, containing parallel corridors that allows to test re-planed paths. After selecting the base and goal positions, the first step of our system is to find the shortest path that minimize the number of nodes, and consequently, the number of robots used. Fig.3 shows the path planned using the Equation 4, resulting in the use of three robots. During the initial deployment strategy, as soon as the robot finds an obstacle not computed initially, a new path is calculated considering the updated visibility graph.

A. Nominal Inward Traversal

This simulation validates the system's ability to successfully deploy a multi-robot communication chain from a base to a target destination. The fleet consists of four robots that establish the communication links while maintaining line-of-sight (LOS) connectivity.

The execution of the generated plan in the simulated environment proceeded without interruption. Connectivity was maintained throughout the entire maneuver, confirming the validity of our deployment methodology. Figure 4 illustrates the arrival of a robot at the point prior to the signal blocking and the subsequent corrective plan.

B. Interrupted Movement with Reactive Replanning

When the lead robot detects the blocked edge, our planner generates a recovery plan. The objective formation for this plan is depicted in purple in Fig. 5. Our methodology describes each step required to reach this formation without losing line-of-sight. In the initial phase shown in Fig. 5, Robot 1 moves first, then Robot 2, and finally Robot 4. Once the new formation is achieved, the fleet will circumnavigate the obstacle. The execution of the remainder of the plan will be detailed in Section V-C.

C. Coordinated Outward Traversal

Following the reconfiguration of the fleet, the coordinated outward traversal is executed as detailed in this section. Fig. 6 illustrates three key snapshots of this process. In each snapshot, the arrows indicate the next intended movement for each robot in the sequence. The captions for each subfigure present the order in which the motions happen, showing the step-by-step execution where the robots move in a coordinated fashion to circumnavigate the signal blockage while maintaining line-of-sight connectivity. At the conclusion of this maneuver, the path to the goal becomes clear for the lead robot, as shown in Fig. 7. For this scenario, considering the start from the base to the obstacle and then the adaptation to the new path to reach the goal, the results in Table I show a gain of 12.21%, 13.25%, and 22.73% in the total distance traveled by all the robots, total r_1 distance traveled and the number of movements made, respectively when comparing our deterministic deployment strategy and the reactive re-planning mechanism proposed.

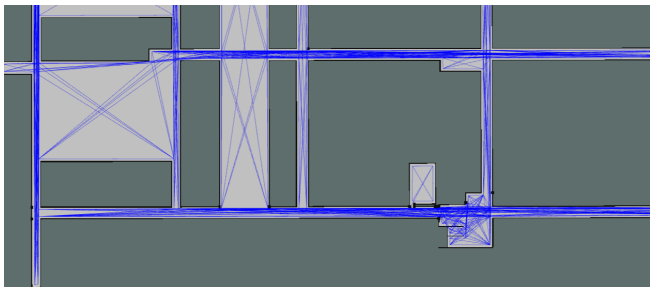
	Deterministic	Adaptative
Total distance traveled (m)	887.2	**778.9**
r_1 distance traveled (m)	373.5	**324.0**
Nº of movements	22	**17**

TABLE I: Comparison between the deterministic deployment strategy and and the adaptive re-planning logic after an obstacle is found.

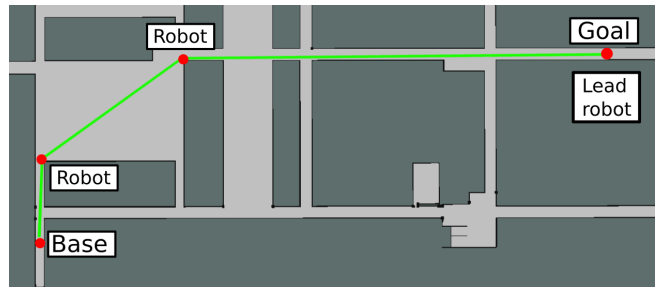
VI. Conclusion

This paper presented a centralized, coordinated motion planner for multi-robot teams, designed to ensure uninterrupted line-of-sight (LOS) connectivity while deploying a communication chain. By leveraging a visibility graph and a custom path-finding algorithm that penalizes the number of required nodes, our methodology first generates an optimal static path for the fleet. We introduced a deterministic deployment strategy and a reactive replanning mechanism to handle environmental interruptions, such as signal blockages. So, it is possible to choose which of the two methods yields the best results to address to the fleet.

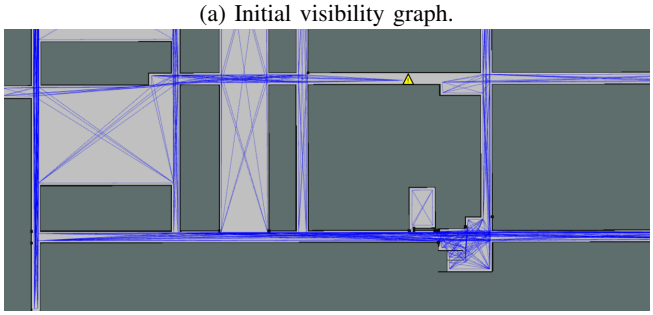
The approach was validated in a simulated environment using ROS 2, Gazebo, and Open-RMF. The results demonstrate that the fleet can: 1) execute a nominal deployment plan, establishing a communication backbone from a base to a target; 2) detect interruptions and reactively generate a recovery plan to reconfigure its formation; and 3) execute a coordinated outward traversal to circumnavigate obstacles while maintaining LOS connectivity at every step. The simulations confirm the viability of our approach for deploying communication networks in environments with unknown obstacles. Future work will focus on extending this



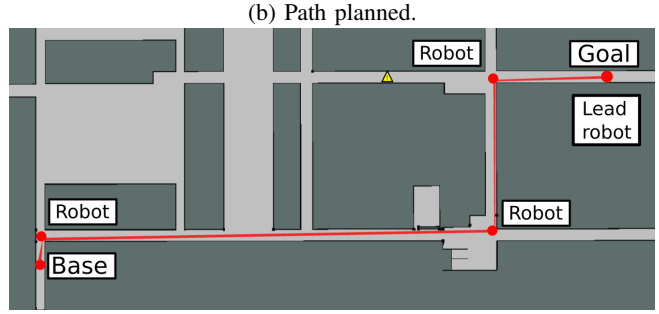
(a) Initial visibility graph.



(b) Path planned.



(c) Updated visibility graph with obstacle (yellow sign).



(d) New path with updated visibility graph.

Fig. 3: Process of deployment process. In Fig.3a, the visibility graph is computed from the given occupancy grid. Then, in Fig.3b, a path that minimizes the distance and number of robots is calculated resulting in the position for each robot to allow the lead robot to reach the goal. occupied by a robot and the goal. As soon as the lead robot found an obstacle, the visibility graph is updated (Fig.3c) and a new path is calculated (Fig.3d) considering it and addressing the robots that will be used.

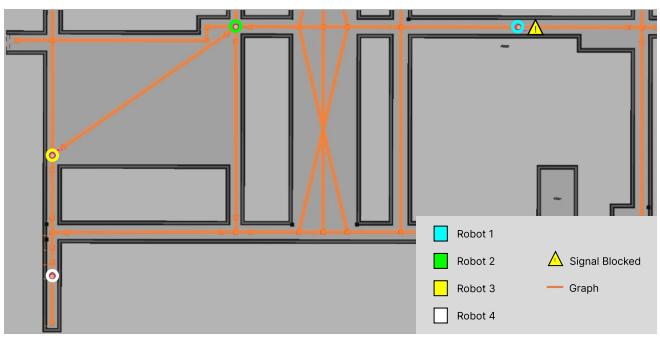


Fig. 4: Result of the nominal inward behavior, where the fleet successfully navigated the graph while maintaining line-of-sight connectivity. The leader halts as intended upon reaching the signal block zone.

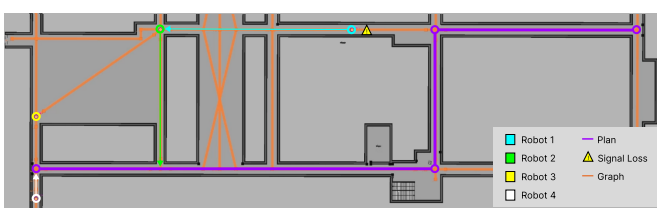


Fig. 5: The recovery plan (purple) generated after detecting the blocked edge. Arrows indicate the initial movements to form a new line-of-sight formation.

methodology to handle dynamic or unknown environments by integrating updates to the visibility graph.

As future work, we aim to improve applicability in communication-denied or intermittently connected environments. Promising directions include hierarchical or distributed variants that tolerate bounded disconnection with guaranteed reconnection, peer-to-peer map fusion and partial consensus when the uplink is unavailable, opportunistic relay formation using idle agents to maintain or restore LoS, and robustness extensions (e.g., local fail-safe controllers and limited-horizon replanning) that ensure safety and task progress during outages. These extensions would preserve the core safety guarantees of the current approach while broadening its operational envelope beyond continuous central connectivity and allowing for further validation of the system with real robots in real-world conditions.

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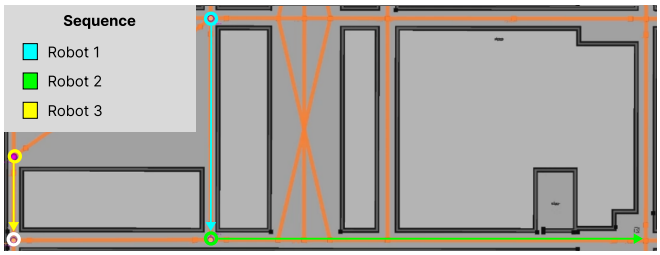
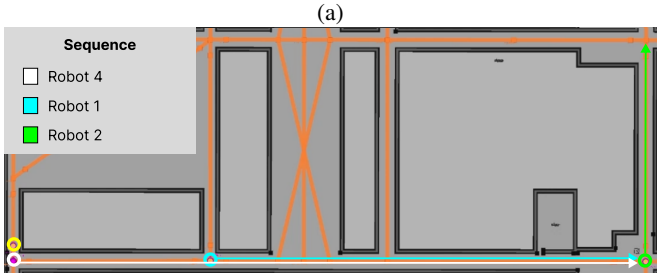
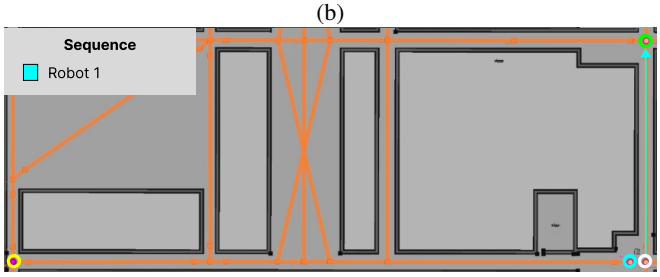


(a)



(b)



(c)

Fig. 6: Snapshots of the coordinated outward traversal. The fleet executes a sequential maneuver around the signal blockage, maintaining line-of-sight throughout the process. Each subfigure shows a step in the sequence, with arrows indicating the next robot to move.

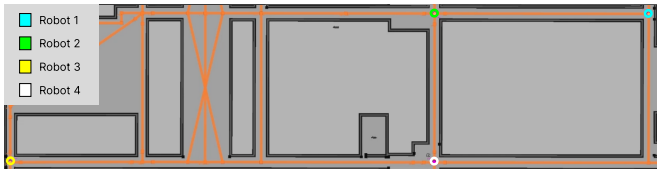


Fig. 7: Final state after the outward traversal, showing a clear, unobstructed path for the lead robot to reach the goal.

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